Business/General Aviation Track #1

The Intelligent FBO: Ready for the Big Game?
Help business aviation prepare for Super Bowl LXI in Los Angeles
Thousands of decisions happen on the ground before an aircraft can arrive, park, refuel, and depart. Major events make that planning more challenging.
Your mission: Build an AI-powered tool that helps an FBO at an LA-area airport prepare for demand during Super Bowl week.
Choose one area:

- **Aircraft arrivals:** When should the FBO expect its busiest periods?
- **Parking:** When might demand exceed available aircraft parking space?
- **Fuel:** How much fuel should the FBO plan to have available?
- **Staffing:** When will additional personnel be needed?

Choose one FBO or a fictional FBO at a real airport. Model a seven-day planning window around the game.
Your prototype should show:

1. An ordinary-week baseline.
2. A forecast for Super Bowl week.
3. A recommended preparation plan.
4. How that plan changes under low, medium, and high demand.

You may use public or synthetic data. Clearly identify assumptions about aircraft arrivals, fuel purchases, parking duration, and available resources. Event attendance alone does not establish aircraft demand.
Measure your impact: Demonstrate forecast accuracy or estimate improvements in resource availability, staffing coverage, parking capacity, or service delays.
Keep humans in command: Protect passenger privacy and explain uncertainty. Your tool supports planning; it does not authorize aircraft movements or replace airport safety procedures.
Stretch goal: Adapt your solution to a selected period during the LA 2028 Olympics.
Be ready to answer: What decision should your AI never be allowed to make?

Commercial Aviation Track #2

Every Gallon Counts: Help Cloudhopper Airways Fly Lighter
Focus Track: AI for Sustainable Airline Operations
Meet Your Airline
Cloudhopper Airways is a fictional global airline connecting destinations through short-, medium-, and long-haul flights.
Your mission: help Cloudhopper carry the supplies passengers need while reducing unnecessary weight, fuel consumption, operating costs, and emissions.
The Challenge
Build an AI-powered planning tool that predicts demand for **potable water or catering supplies** and recommends appropriate loading quantities.
Find the balance between reducing excess supplies and maintaining passenger service, required minimums, and a reasonable buffer for unexpected changes.
Your 30-Hour Scope
Choose:

- **One supply category:** potable water or catering supplies.
- **One aircraft type.**
- **Three representative routes** from Cloudhopper's fictional global network.

Build one useful workflow for an airline operations planner.
What Could You Build?
Create a prototype that lets a planner select a flight and see:

- Predicted supply demand based on passenger count, flight duration, departure time, and historical consumption.
- Recommended loading quantities, including a clearly explained buffer.
- A comparison with Cloudhopper's standard loading practice.
- Estimated weight reduction and potential fuel, cost, and CO_2 savings.
- How the recommendation changes when passenger numbers increase or a flight is delayed.

Data and Assumptions
Use a shared synthetic starter dataset containing:

- Flight schedules and route distances.
- Aircraft type and passenger capacity.
- Expected and actual passenger counts.
- Historical supply consumption.
- Standard loading quantities and minimum supply requirements.
- Simplified assumptions for translating weight reduction into fuel, cost, and emissions savings.

Clearly identify fictional data, assumptions, and limitations. Treat calculated savings as estimates rather than proven airline results.

Measure Your Impact

Compare your solution against Cloudhopper's standard loading practice. Show:

- Prediction accuracy on data held back from model development.
- Excess supply weight reduced.
- Whether passenger demand and minimum supply requirements were met.
- Estimated fuel, operating cost, and CO_2 savings.
- Performance during at least one disruption scenario.

Explain why your AI approach is useful compared with a simple rule or historical average.

Responsible AI and Human Oversight

Airline staff must approve loading decisions. Your tool must:

- Respect the provided minimum supply requirements.
- Explain recommendations and uncertainty.
- Include a buffer for changing conditions.
- Use aggregate passenger information without sensitive personal data.

Required aircraft fuel reserves remain outside this challenge's optimization scope.

Be ready to answer: What decision should your AI never be allowed to make?

What to Deliver

1. **Working prototype:** Demonstrate one complete planning workflow.
2. **Three-minute pitch:** Explain the user, problem, solution, and value.
3. **Impact comparison:** Show your baseline, results, assumptions, and limitations.
4. **Disruption demonstration:** Test increased passenger demand or a flight delay.
5. **Responsible AI statement:** Identify safeguards and decisions that remain with humans.

Your Mission in One Sentence

Help Cloudhopper Airways load what passengers need, reduce unnecessary weight, and demonstrate how smarter supply planning could save fuel—one flight at a time.